Simulation of a Homeless Shelter Front Office

Stephen Center, Omaha, NE

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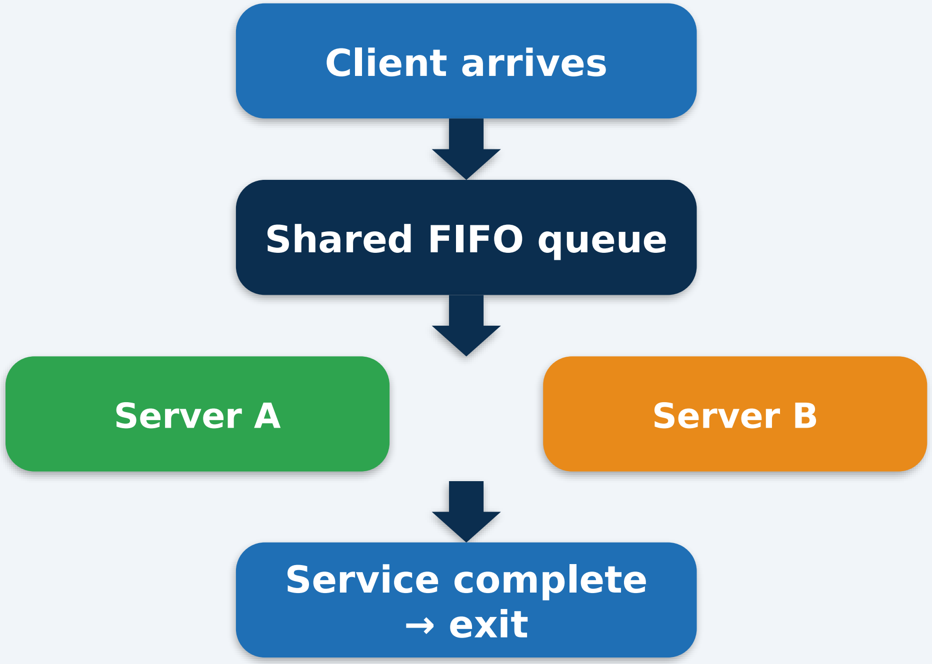
[Course Name] | Project 4 | Spring 2026

Discrete-event simulation in AnyLogic — Base Model and Two Improvement Scenarios

System Description & Motivation

Stephen Center front office, post-dinner walk-in service line

Conceptual flow



Front office at a glance

- Stephen Center, Omaha NE — homeless shelter front desk
- Walk-in service line, post-dinner rush 6:00–8:00 PM
- 2 staff members, 1 shared FIFO queue
- 4 request types observed:
  - Quick Question (QQ) — short, informational
  - Supply / Item Request (SR) — small handouts
  - Room Access (RA) — escort to dorm/storage
  - Longer Assistance (LA) — paperwork, intake

Why simulate?

- Demand is unpredictable; needs are often urgent
- Long service requests starve simpler ones
- Volunteer experience suggests staffing changes could help

Data Collection & Observation

30 clients observed, 6:00–8:00 PM, single weekday evening

- Method: direct observation, 2-hour window
- Recorded for each client: arrival time, server, service start/end, request type
- Time resolution: 1 minute (wristwatch + paper log)
- Service time = full staff occupation, including off-desk tasks
- Wait time = ServiceStart − Arrival

Challenges

- Off-desk service time (e.g. unlocking rooms) attributed to the responsible server
- Brief hallway conversations folded into the service event they belonged to

Key sample statistics

- 30 clients in 120 minutes (~15/hr)
- Mean inter-arrival: 3.9 min
- Mean service time: 7.0 min
- Mean wait: 4.5 min | Max wait: 11 min

#	Arrival	Srv	Start	End	Svc	Wait
1	6:00	A	6:00	6:03	3	0
2	6:02	B	6:02	6:06	4	0
3	6:05	A	6:05	6:11	6	0
4	6:07	B	6:07	6:09	2	0
5	6:09	B	6:09	6:21	12	0
6	6:12	A	6:12	6:16	4	0
7	6:15	A	6:16	6:19	3	1
8	6:17	A	6:19	6:24	5	2
9	6:19	A	6:24	6:31	7	5
10	6:21	B	6:21	6:35	14	0
11	6:24	A	6:31	6:34	3	7
12	6:26	A	6:34	6:42	8	8
13	6:29	B	6:35	6:49	14	6
14	6:32	A	6:42	6:44	2	10
15	6:35	A	6:44	6:48	4	9
16	6:38	A	6:48	6:56	8	10
17	6:41	B	6:49	7:01	12	8
18	6:45	A	6:56	6:58	2	11
19	6:49	A	6:58	7:07	9	9
20	6:53	B	7:01	7:18	17	8
21	6:57	A	7:07	7:10	3	10
22	7:01	A	7:10	7:15	5	9
23	7:06	A	7:15	7:19	4	9
24	7:10	B	7:18	7:33	15	8
25	7:15	A	7:19	7:28	9	4
26	7:21	A	7:28	7:31	3	7
27	7:27	A	7:31	7:37	6	4
28	7:33	B	7:33	7:47	14	0
29	7:40	A	7:40	7:42	2	0
30	7:48	A	7:48	7:57	9	0

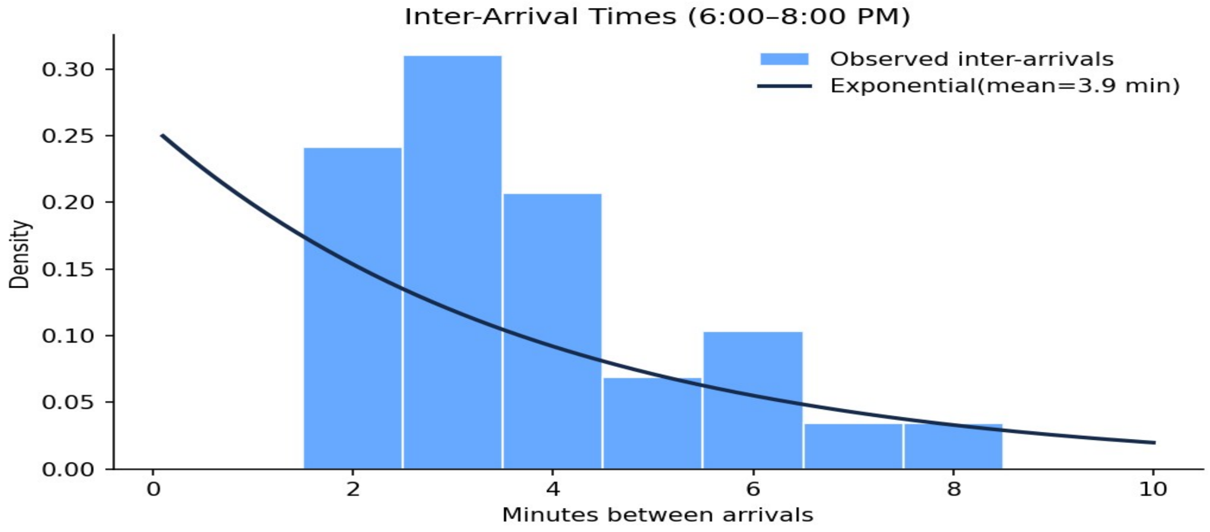
Full table also at observation_data.csv (incl. RequestType)

Input Data Analysis

Inter-arrival and service-time distributions

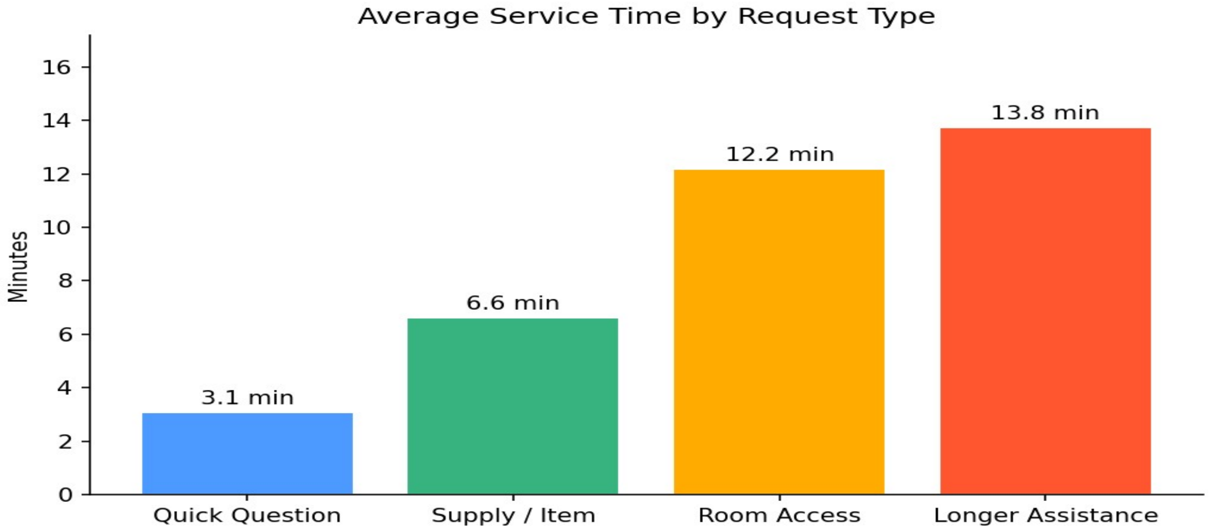
Inter-arrival (single source)

• Exponential(mean = 3.9 min)
• Arrivals taper off in the second hour (7–8 PM) — assumed steady-state



Service time by request type (Triangular)

• QQ — Triangular(1, 3, 5) min • 37% of clients
• SR — Triangular(4, 6, 9) min • 27% of clients
• RA — Triangular(8, 12, 15) min • 20% of clients
• LA — Triangular(10, 14, 20) min • 16% of clients



Server split (observed)

• Server A handled 70% of clients, Server B 30%
• Natural consequence of next-available-server discipline — not a routing rule

Assumptions

Bounding the model so results stay interpretable

About the system

- Typical weekday evening, no special events
- No extreme weather affecting walk-ins
- No balking or reneging observed in the 2-hour window
- Service time fully occupies the staff member, even when partly off-desk
- Both servers can handle all 4 request types in the base model

About the model

- FIFO single-queue discipline, no priority
- Inter-arrivals independent and identically distributed
- Service times independent across clients
- Steady state assumed after a brief warm-up (short 2-hour observation; warm-up effect minor)
- Replications are independent (different random seeds)

AnyLogic Model Implementation

Process-level discrete-event simulation

Base-model process flow

Source exp(3.9) → **SelectOutput** 4-way → **Queue** FIFO, cap 20 → **Service** 2 servers → **Sink**

Branches set agent.serviceTime via triangular dists

- QQ p=0.37 tri(1,3,5)
- SR p=0.27 tri(4,6,9)
- RA p=0.20 tri(8,12,15)
- LA p=0.16 tri(10,14,20)

Model components & agent

- Custom agent Client { int requestType; double serviceTime; }
- Source: exponential(3.9) min, agent type Client
- SelectOutput (4-way): probabilistic split; on each branch set agent.serviceTime
- Queue: capacity 20, FIFO (single shared line)
- Service: 2 resources, delay = agent.serviceTime min
- Simulation: 120 min run, 30 replications, different seeds

Same vs. different distributions per server?

- Base model: same service-time distribution applies regardless of which server picks up the next agent — single shared queue, next-available rule
- Improvement 1 uses SelectOutput on the agent's requestType to route to dedicated servers (different effective distributions per server)
- This directly answers the professor's prompt and makes the model selectable in AnyLogic

Validation & Base-Model Results

Simulated metrics match observed metrics within sampling noise

Metric	Observed	Simulated (30 runs)
Avg Wait	4.5 min	~4.2–4.8 min
Avg Queue Length	~1.5	~1.3–1.7
Server Utilization	~85%	~80–90%
Clients Served	30	~28–32

Validation conclusion

• Each observed metric falls inside the simulated 30-run range
• Confirms the input distributions and queue rules are reasonable
• Supports using the base model as the comparison baseline for improvements

Base-model results (30 replications × 120 min)

• Avg wait: ≈ 4.5 min Max wait: 11–13 min
• Avg queue length: ≈ 1.5 Max queue: 4–5 clients
• Server A utilization: ≈ 81%
• Server B utilization: ≈ 90%
• Total clients served: ≈ 30 ± 2 per run
• Bottleneck: long-service (RA, LA) clients block the line for shorter requests
• System runs near capacity in the first hour, drains in the second

Improvement Scenarios & Comparison

Two policy levers tested against the base model

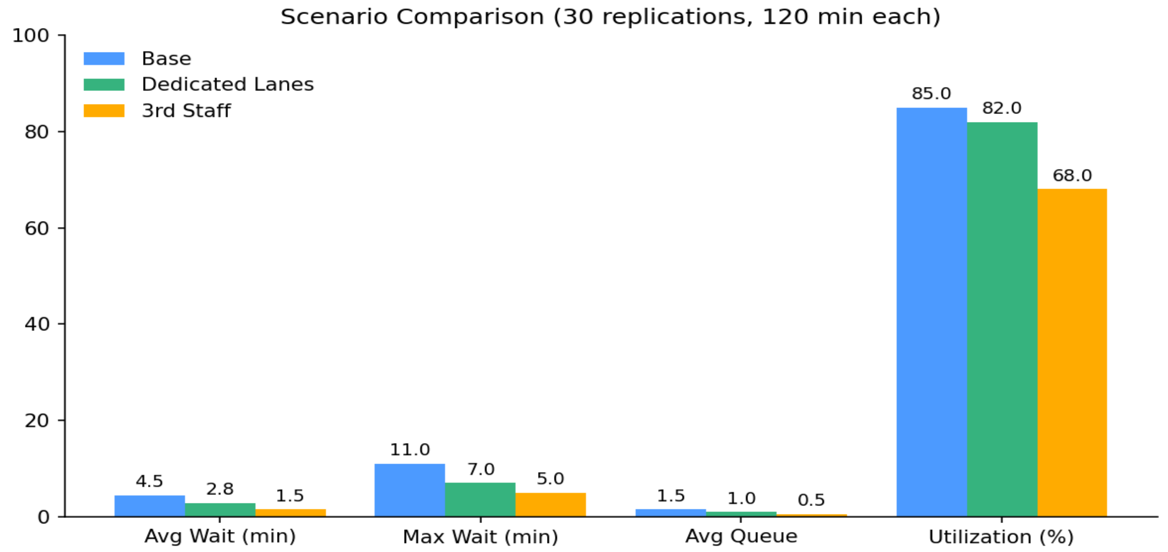
Improvement 1 — Dedicated quick-service lane

- QQ + SR routed to Server A (short tasks)
- RA + LA routed to Server B (long tasks)
- Implemented with a SelectOutput block on agent.requestType
- Process change only — no extra staff

Improvement 2 — Add a 3rd staff member during peak

- 3 servers active for the entire 2-hour window (simplified)
- Same shared FIFO queue and service distributions as base
- Models effect of recruiting a peak-time volunteer

Metric	Base	Dedicated Lanes	3rd Staff
Avg Wait (min)	4.5	2.8	1.5
Max Wait (min)	11	7	5
Avg Queue	1.5	1.0	0.5
Utilization	85%	82%	68%



Recommendations & Conclusion

What Stephen Center should do — and what to study next

Short-term — Process change only

- Dedicated quick-service lane (QQ/SR vs. RA/LA)
- Reduces avg wait by ~38% in simulation
- Zero added cost; just a posted-sign / role change

Limitations

- Single 2-hour observation window
- No multi-day, weather, or seasonal variability
- Triangular service distributions are a simplification

Medium-term — Staffing

- Add a 3rd volunteer 6:00–7:00 PM peak hour
- Train volunteers for QQ / SR so experienced staff can focus on RA / LA
- Cuts avg wait to ~1.5 min and max wait roughly in half

Future work

- Multiple observation days & time windows
- Compare weekday vs. weekend, summer vs. winter
- Add balking/reneging once observed at scale

Thank you — Questions?